WEEK 1 · DAY 2

AWS Global Infrastructure

BaseStack Academy · AWS Cloud Accelerator

Cloud Foundations

Today's Agenda

AWS Global Infrastructure

1	Overview — Why This Matters	FOUNDATION
2	Key Concepts — Definitions & Vocabulary	CORE KNOWLEDGE
3	Deep Dive — Key Concepts Extended	DEEPER LEARNING
4	How It Works — Step by Step	MECHANICS
5	Technical Deep Dive	ADVANCED DETAIL
6	Architecture & Visual Model	VISUAL
7	Real-World Nigerian Scenario	CONTEXT
8	Common Mistakes & Exam Traps	EXAM PREP

Overview — Why This Matters

AWS operates one of the largest and most reliable cloud infrastructures in the world, spanning 39 geographic Regions and 123 Availability Zones as of 2026 — with continuous expansion. Understanding how this infrastructure is organised is critical for designing systems that are resilient, performant, and compliant with data sovereignty.

For Nigerian businesses, the nearest AWS Region is af-south-1 in Cape Town, South Africa. Understanding latency implications, data residency laws (critical for fintech and healthcare), and disaster recovery across regions will become increasingly relevant as Africa's digital infrastructure matures.

WHY NOW?

Cloud skills are the #1 most in-demand tech capability globally. Understanding this topic is essential for both the SAA-C03 exam and real cloud engineering work.

Beginner Note

Do not worry if this feels new — every expert was once a complete beginner. Take notes and ask questions freely.

Key Concepts — Know These Definitions

Study tip: Say each definition aloud — spoken repetition builds memory 3× faster than reading alone.

AWS Region

A geographic area containing 2+ Availability Zones, physically isolated from other Regions

Availability Zone (AZ)

One or more discrete data centres with independent power, cooling, and networking within a Region

Edge Location

A smaller AWS facility for caching content close to end users — used by CloudFront CDN and Route 53 DNS

Local Zone

An extension of an AWS Region that places compute and storage closer to specific metropolitan areas

Wavelength Zone

AWS infrastructure embedded directly in telecom networks for ultra-low latency 5G applications

Key Concepts (Continued)

Concepts Summary

All key concepts for this lesson have been covered. Use this slide to review and quiz yourself.

- ✓ AWS Region
- ✓ Availability Zone (AZ)
- ✓ Edge Location
- ✓ Local Zone
- ✓ Wavelength Zone

⚙️ How It Works — Step by Step

1. When you choose a Region for your application, you select a geographic boundary within which your data will primarily reside. Each Region contains at least two AZs — typically three or more — that are physically separated by tens of miles. They are far enough apart that a single natural disaster (flood, earthquake, or massive power failure) will not take down more than one AZ at the same time, yet they are close enough to be connected by ultra-low latency private fibre networks.

2. Edge Locations (also called Points of Presence or PoPs) are distributed even more widely — 400+ globally. They are used by Amazon CloudFront to cache static content (images, videos, CSS files) close to end users so pages load faster. When a user in Abuja requests a webpage hosted in us-east-1, CloudFront may serve the cached content from a nearby Edge Location in Lagos, drastically reducing latency and loading the page almost instantly.

3. You cannot choose which specific data centres your data goes to within an AZ. AWS manages that physical allocation. What you can control is the Region and AZ.

⚙️ Technical Deep Dive

Understanding the Fundamentals

Problem: Single points of failure destroy businesses.
Solution (Regions & AZs): Data lives in a geographic Region spread across 2+ physically isolated Availability Zones. If one AZ fails due to a disaster, the other takes over instantly via private fiber networks.

Problem: Long physical distances cause high latency (slow buffering).
Solution (Edge Locations): 400+ global Points of Presence use Amazon CloudFront to cache static content right around the corner from the end user, ensuring pages load in milliseconds.

⍰ Architecture & Visual Model

Core Components

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How They Connect

Local Zone

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Real-World Nigerian Scenario

Adaeze builds a video streaming platform for Nollywood content. She deploys her application in the af-south-1 Cape Town Region across two Availability Zones — AZ-1 (af-south-1a) and AZ-2 (af-south-1b). If an electrical transformer failure takes down AZ-1, her load balancer immediately routes all traffic to AZ-2 — zero downtime for users. She also enables CloudFront distribution for her video files, so a viewer in Kano loads video from the nearest Edge Location rather than Cape Town

Discussion Question

Think about a Nigerian business you know. How would this AWS service change how they operate? What problem would it solve?

⚠️ Common Mistakes & Exam Traps

● These are the most common exam errors — every one costs marks. Memorise them.

✗ AZs within a Region are connected via LOW-LATENCY PRIVATE fibre — never the public internet. This private connection is what makes Multi-AZ failover fast and reliable.

✗ Edge Locations are NOT Availability Zones. They handle caching and DNS only — you cannot launch EC2 instances or RDS databases in Edge Locations.

✗ Data does NOT automatically leave a Region. Cross-Region replication must be explicitly configured. This is critical for data sovereignty compliance.

SAA-C03 Practice Question

Question

A Solutions Architect at a Lagos fintech company is designing the AWS infrastructure for a new payment platform. They need to implement best practices related to AWS Global Infrastructure. Which of the following statements is CORRECT?

A Data stored within a specific AWS Region is not replicated to other Regions automatically unless explicitly configured by the architect.

B An AWS Region consists of a single, massive data centre that serves an entire geographic area.

C Availability Zones within a Region are connected to each other over the public internet to ensure maximum global reach.

D Edge Locations are specialized Availability Zones where the architect can deploy primary EC2 database servers for localized compute.

SAA-C03 Practice Question

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Key Takeaways — What You Learned Today

AWS Global Infrastructure

Concepts Covered

- ✓ AWS Region
- ✓ Availability Zone (AZ)
- ✓ Edge Location
- ✓ Local Zone
- ✓ Wavelength Zone

Before Next Class

1. Review the concepts above without notes
2. Complete the hands-on console lab
3. Post your progress on LinkedIn / X
4. Review the exam traps from the previous slide
5. Check the Further Resources list

Further Resources & Reading

Go deeper with these official AWS resources:

https://en.wikipedia.org/wiki/Nigeria_Data_Protection_Commission

https://aws.amazon.com/about-aws/global-infrastructure/

Homework: Read at least ONE link above before the next class. Post what you learned on the WhatsApp group.

BASE STACK

A C A D E M Y

Week 1 · Day 2 Complete

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Cloud Foundations

Next: Hands-on Console Lab →